

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11: GROWTH AND DEVELOPMENT IN PLANTS

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the driving question: What causes the differences in germination, emergence and growth of maize and bean plants? Think back to Mr. Wafula's farm near Kitale. He planted maize H614 and Rosecoco beans on the same day (Day 0), 3 cm deep, after 25 mm of rain. Use the evidence you collected in Lessons 1 to 10. Write your answers in the space under each prompt. Use full sentences, include numbers from the data, and use the words germination, emergence, dormancy, viable, primary growth and hormones correctly. Where a prompt asks for Claim, Evidence and Reasoning, write all three parts. Where a data table is given, use it. Work on your own.

Part 1: Seed structure and emergence (CER)

Think about the time-lapse videos of the bean and maize seedlings that you watched. Write a Claim-Evidence-Reasoning explanation of how the different seed structures of maize H614 and Rosecoco beans lead to different ways of emerging. In your answer: (a) name the type of seed each crop has and how many cotyledons it has; (b) say where each crop stores its food; (c) use what you saw in the time-lapse videos to say what happens to the cotyledon(s) of each crop during germination and what this type of germination is called, and give the day each crop emerged above the soil; (d) explain the difference between germination and emergence.

Write your answer here:

Part 2: Why healthy-looking seeds did not germinate

Use the table from Mr. Wafula's field at Day 14 (100 seeds of each crop).

Reason seed did not germinate	Maize H614	Bean Rosecoco
Eaten or rotted by fungi/pests	5	2
Old seed with dead embryo (not viable)	4	4
Waterlogged low patch (no oxygen)	3	6
Hard impermeable seed coat	0	8
Total that did not germinate	12	20

Write a CER explanation of why 12 maize and 20 bean seeds did not germinate. In your answer: (a) say which seeds were dormant and what dormancy means; (b) the 8 hard-coated bean seeds were later scarified and replanted as a separate re-test after Day 14. Say what happened to them and what this shows about the cause of their dormancy; (c) explain why the other failed seeds were not dormant; (d) name the hormone that maintains dormancy and the hormone that helps germination.

Write your answer here:

Part 3: Conditions needed for germination

Fatuma set up germination jars with 20 Rosecoco bean seeds in each jar. These are the Day 5 results.

Jar	Conditions	Seeds germinated (out of 20)
A	Water + air + 25 C	18
B	Dry	0
C	No air (submerged in boiled, cooled water with oil layer)	1
D	Cold (fridge, 4 C)	0

Write a CER explanation of which conditions a viable, non-dormant seed needs to germinate. Then use the table, and the numbers of seeds that failed in the waterlogged low patch of Mr. Wafula's field, to explain what may have happened there. Say how the jar and the field are alike and how they are different.

Write your answer here:

Part 4: Growth after emergence

After emergence the two crops grew differently. Use this data from Mr. Wafula's field (mean shoot height of 10 seedlings, in cm).

Day	7	14	21	28	35	42
Maize H614	3	12	30	62	98	125
Bean Rosecoco	4	9	18	30	40	44

Stem width at Day 14, 28 and 42 was 6, 14 and 22 mm for maize and 2, 4 and 6 mm for bean. Write a CER explanation of how the growth of maize and bean differ. In your answer: (a) say between which days each crop grew the most and how many centimetres it grew in that period; (b) say what shape a growth curve has and name the three phases, then say which phases you can see in the bean data and in the maize data up to Day 42; (c) explain what primary growth is and where it happens; (d) explain why maize gets thicker even though it does not make wood.

Write your answer here:

Part 5: Hormones, apical dominance and your final model

Mr. Wafula compared two groups of Rosecoco bean plants. In one group the shoot tip was left intact; in the other the shoot tip was removed at Day 20. Mr. Wafula counted the branches on each group at Day 25 and Day 32. (a) Using what you learned about hormones, write a CER explanation of what you would expect the two groups to look like and why. Use the numbers of branches from Lesson 7. (b) Name the hormone made in the shoot tip that affects lateral buds, say what it does to them, and name the hormone that promotes lateral buds. (c) Now write a short final answer to the driving question: What causes the differences in germination, emergence and growth of maize and bean plants? Finish by saying two ways in which your final model is different from your Day 1 drawing.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy	All terms (germination, emergence, dormancy, viable, primary growth, hypogeal, epigeal, hormones) are used correctly, with no errors, and the maize/bean differences are right.	Most terms and ideas are correct, with one or two small errors that do not change the meaning.	Several ideas are wrong or terms are mixed up, for example confusing germination with emergence or saying maize makes wood.
Use of evidence	Uses exact data from Mr. Wafula's farm in every section, such as 88 and 80 germinated, 8 dormant beans with 7 germinating, the jar results, heights and branches.	Uses some data from the unit, but misses a part or quotes a number loosely.	Gives few or no numbers from the unit, or the numbers do not match the data.
Reasoning	Clearly links each piece of evidence to a	Links most evidence to ideas, but some	Lists facts without explaining how they

	biological idea, explaining why it supports the claim, such as water and enzymes, oxygen and respiration, and auxin and lateral buds.	links are brief or incomplete.	support the claim.
Connection to the driving question and model revision	Gives a complete answer to the driving question covering seed structure, seed condition, environment and hormones, and says two clear changes from the Day 1 model.	Answers the driving question with most causes and gives at least one change from the Day 1 model.	The answer covers only one cause or does not say how the model changed.
Clarity of communication	Well organised Claim, Evidence and Reasoning in clear sentences, with data read correctly from the tables.	Mostly clear and organised, with a few unclear sentences.	Hard to follow, with the Claim, Evidence or Reasoning missing.

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